



HandVerify

A comparative study of deep learning architectures for writer verification
6 backbones x 3 losses x 4 dataset pairings

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Handwriting as a biometric trait

- **Behavioral** biometric: a learned motor skill, not a fixed physiological trait.
- Strong **inter-writer** distinctiveness, but non-trivial **intra-writer** variability.
- Uses: forensic document examination, signature authentication, manuscript attribution.

Writer verification

Deciding whether two handwriting samples share an author: both valuable and challenging.

What makes it hard

Intra-writer variability

same writer, different letter size, stroke pressure and legibility

Multi-individuality

some writers maintain two or more distinct styles

Look-alike writers

different writers sharing slant, spacing and regularity

Handcrafted features

degrade under writing variability and noise, transfer poorly across datasets



Verification, not identification

Identification (1:N)

- Probe assigned to one of a **fixed set** of enrolled writers
- Closed set
- Metrics: top-k accuracy, mAP

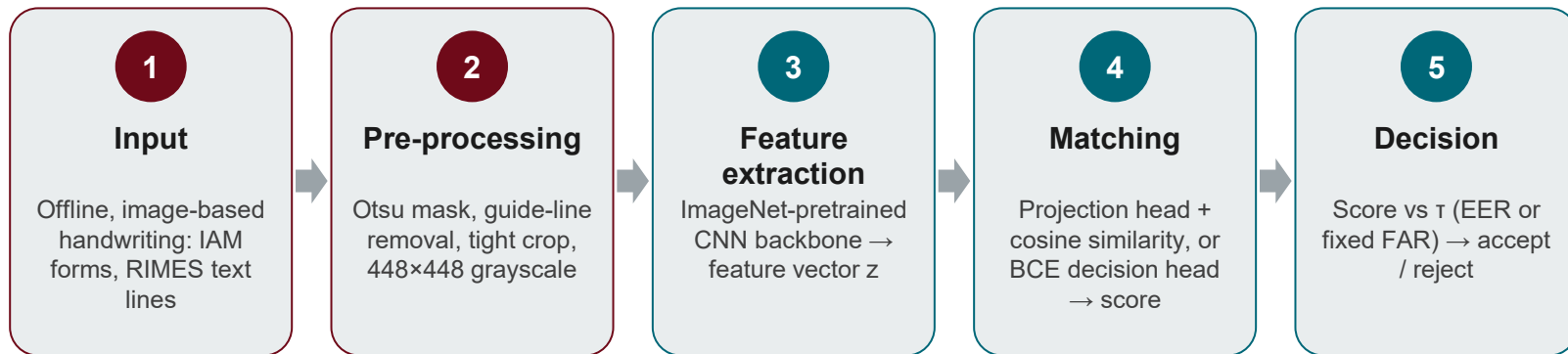
Verification (1:1): (HandVerify)

- “Do these two samples share the **author?**”
- Explicit similarity model + **decision threshold**
- Metrics: EER, AUC, d' , GAR at fixed FAR
- **Writer-independent**: test writers never seen in training

Why a Siamese network? With unseen writers, closed-set identity classification is unsuitable as the decision mechanism. Pairwise and metric-learning formulations learn a similarity instead: a paradigm widely adopted for face and signature verification.



HandVerify as a biometric pipeline

**Verification**

one-to-one comparison between a probe sample and a claimed identity

Shared weights

both images go through the same encoder (shared θ): symmetric feature extraction



Research gap and contributions

72

trained configurations

6 CNN backbones

× 3 training losses

× 4 train → test pairings

Gaps in prior work

- A single backbone **or** a single training objective per study
- Cross-dataset generalization rarely tested
- Loss functions seldom compared under controlled conditions

Our contributions

- 1 Systematic comparison of 3 learning paradigms × 6 CNNs, same- and cross-dataset
- 2 Two training strategies: epoch-wise negative resampling and selective layer freezing
- 3 Biometric evaluation: EER, AUC, decidability and fixed-FAR operating points
- 4 Paired Wilcoxon analysis of loss-function consistency



Datasets: IAM and RIMES

IAM

English · full handwritten forms

657

authors

1,539

processed images

2.34 / 1

mean / median per author

59

max (one outlier author)

Strong right skew: roughly half the writers have a single page

RIMES

French · individual text lines

1,500

authors

5,046

stacked 448×448 images

3.36 / 3

mean / median per author

10

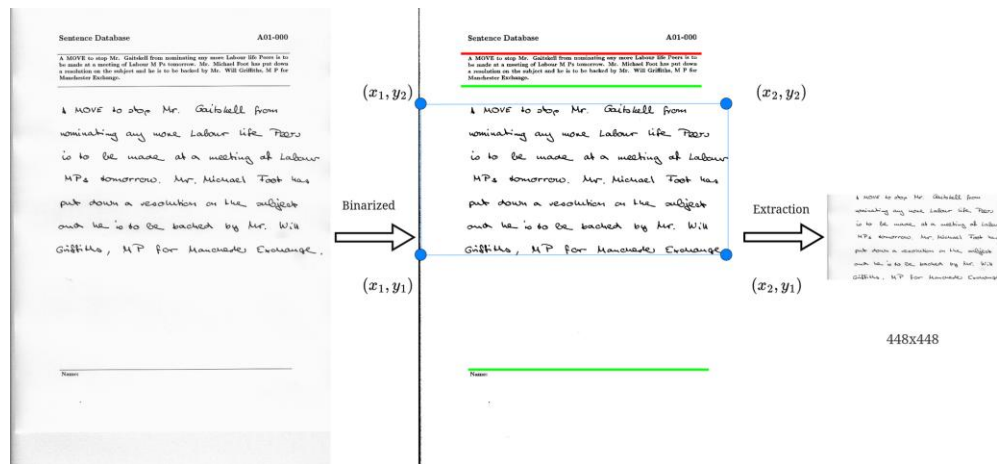
max per author

Broader, more centrally distributed spread

Design consequences: writer-level (not image-level) splits prevent leakage; cross-dataset pairings test a different domain and collection protocol.



Preprocessing IAM forms

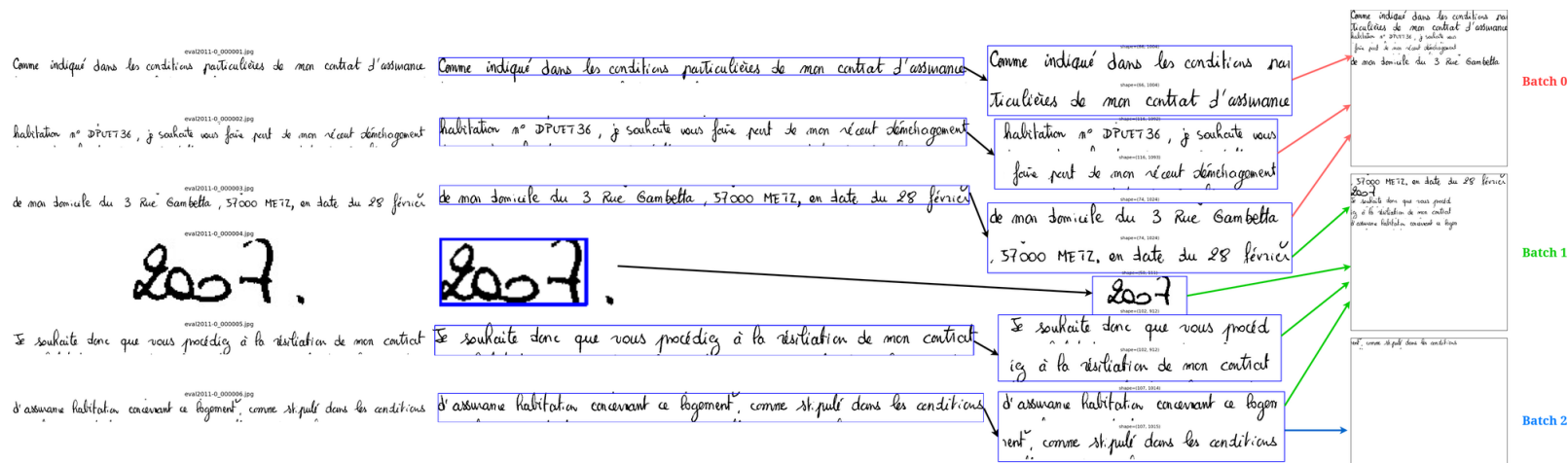


Binarisation as a guide: crop coordinates come from the mask, pixels from the original grayscale; fallbacks avoid empty or inverted regions.

- 1 **Otsu threshold** $t^* = \operatorname{argmax} \sigma_B^2(t)$: binary mask used **only as a guide**
- 2 Guide lines by **morphological opening** (erosion $\rightarrow$ dilation), kernel $\max(30, w/15)$
- 3 Keep region between detected lines (fallback: 22–85 % of height)
- 4 **Tight crop** from row / column ink projections ($\rho = 0.02$), 5-px padding
- 5 Centre on square canvas $\rightarrow 448 \times 448$, area interpolation



Preprocessing RIMES text lines



1. Trim & tight crop

drop rows > 50 % black; projection crop
($\rho = 0.018$)

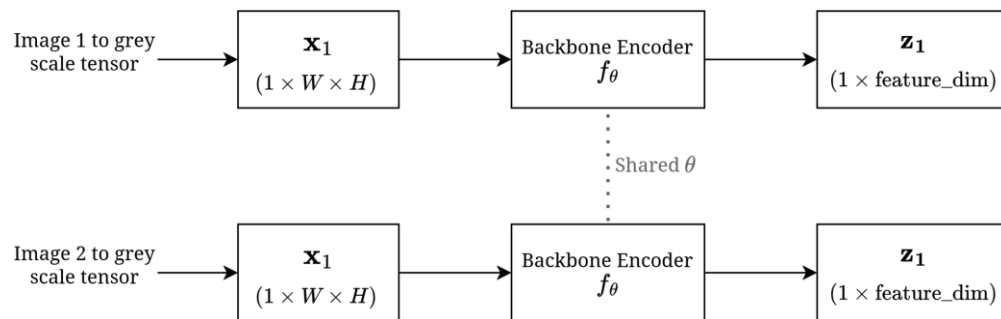
2. Height normalisation

per writer: $h_{\min} = \text{clip}(\min h, 20, 32)$ px,
aspect preserved

3. Stack into canvases

5 lines per 448×448 image; lines > 896
px split in two

Siamese encoder with shared weights



Grayscale adaptation

$$W_{\text{gray}} = \frac{1}{3} (W_R + W_G + W_B)$$

Conv2d(3 → C) becomes Conv2d(1 → C)
without re-initialisation: pretrained filters are kept.

Shared weights

same θ on both branches → symmetric feature extraction

Writer-independent

a pairwise similarity instead of closed-set classification: generalises to unseen writers

Modular

backbone encoder and training objective are swappable: two-axis comparison



Six pretrained CNN backbones

ResNet-18 / 34



Residual connections for deep feature learning

EfficientNet-B0 / B1



Compound scaling: comparable accuracy at a fraction of the parameters

MobileNetV3-Small / Large



Depthwise-separable convolutions for efficiency

ImageNet-pretrained

domain gap between natural images and near-binary handwriting: the best backbone is non-obvious

$L_f = 3$ frozen layers

generic low-level features preserved, deeper layers fine-tuned

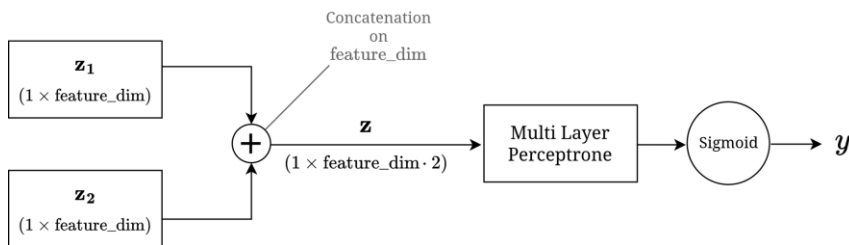
$l_r = 10^{-4} / 5 \cdot 10^{-5}$

the smaller rate for models with > 15 M trainable parameters (AdamW)

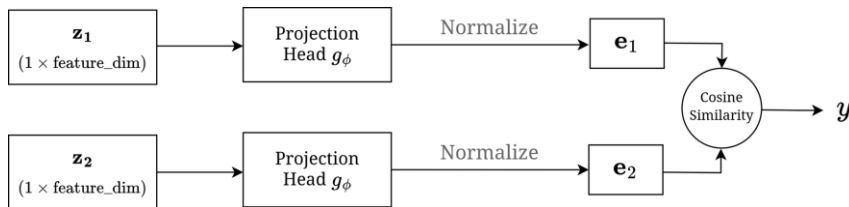


Two heads for two paradigms

BCE: learned decision head



Contrastive / Triplet: projection head



$$[z_1; z_2] \in \mathbb{R}^{2D} \rightarrow 32 \rightarrow 16 \rightarrow 8 \rightarrow 4 \rightarrow 2 \rightarrow 1 \rightarrow \sigma$$

- BatchNorm + ReLU + graduated dropout per layer
- **No explicit geometric constraint** on the embedding space

$$D \rightarrow 1024 \rightarrow 512 \rightarrow 256 \rightarrow 32, \hat{\mathbf{e}} = \mathbf{e} / \|\mathbf{e}\|_2$$

- Score $\mathbf{s} = \hat{\mathbf{e}}_1^T \hat{\mathbf{e}}_2 \in [-1, 1]$, distance $d = 1 - s$
- **Same head** for Contrastive and Triplet: only the objective differs



Three training objectives

BCE

$$L = -[y \log \hat{y} + (1-y) \log(1-\hat{y})]$$

- Direct pairwise **classification**
- Stable gradients, reliable convergence
- Embedding space **not** explicitly structured

Contrastive (m = 1.0)

$$L = y d^2 + (1-y) \max(m - d, 0)^2$$

- Genuine pairs pulled **together**
- Impostors penalised if **closer than m**
- Operates on **pairs** of embeddings

Triplet (m = 0.5)

$$L = \max(d_{ap} - d_{an} + m, 0)$$

- Anchor / positive / negative
- **Relative** constraint: $d_{ap} + m < d_{an}$
- Triplet Loss of **FaceNet** (Schroff et al.)

$d = 1 - \cos(e_1, e_2)$ on L2-normalised embeddings, $d \in [0, 2]$; margins set from preliminary experiments.



Dynamic negative sampling

3,262

genuine training pairs (IAM $\rightarrow$ IAM)

589,154

impostor pair candidates

genuine $O(W \cdot N^2)$ vs impostor $O(W^2 \cdot N^2)$
impostor pairs vastly outnumber genuine pairs

1

Keep **all genuine pairs**

minority class, generated once and kept fixed

2

Build the full **impostor pool**

every different-writer pair is a candidate

3

Each epoch sample $N_{\text{imp}} = N_{\text{gen}} \cdot (1-r)/r$

$r = 0.5 \rightarrow$ balanced, with different negatives every epoch

4

Shuffle and train

validation and test pairs are **never** resampled

Forensic prior: casework is skewed towards disputed, often forged material (ascertainment bias): train on diverse negatives.



One training recipe for all 72 models

FREEZING

3

first backbone layers frozen:
pretrained low-level features kept

DROPOUT

0.20 → 0.12

graduated: deeper layers retain more
expressiveness

OPTIMISER

AdamW

$\beta = (0.9, 0.999)$, decoupled weight
decay 10^{-4}

LR SCHEDULE

×0.5

ReduceLROnPlateau, patience 2, on
validation loss

EARLY STOPPING

5

epochs of patience (max 50):
prevents overfitting

AUGMENTATION

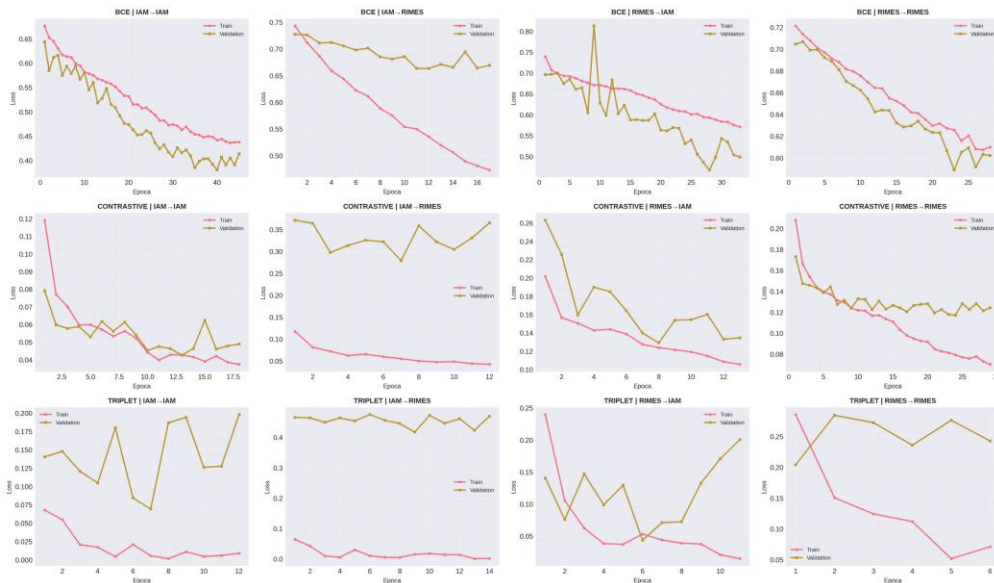
±15°

rotation + resized crop 90–110 %;
batch 16, 448×448



Training dynamics

Training Curves — backbone: resnet18



ResNet-18 · rows: BCE, Contrastive, Triplet · columns: the 4 train → test pairings

Stable optimisation

training loss decreases (nearly) monotonically

BCE

validation tracks training on same-dataset splits; early divergence cross-dataset (e.g. IAM → RIMES)

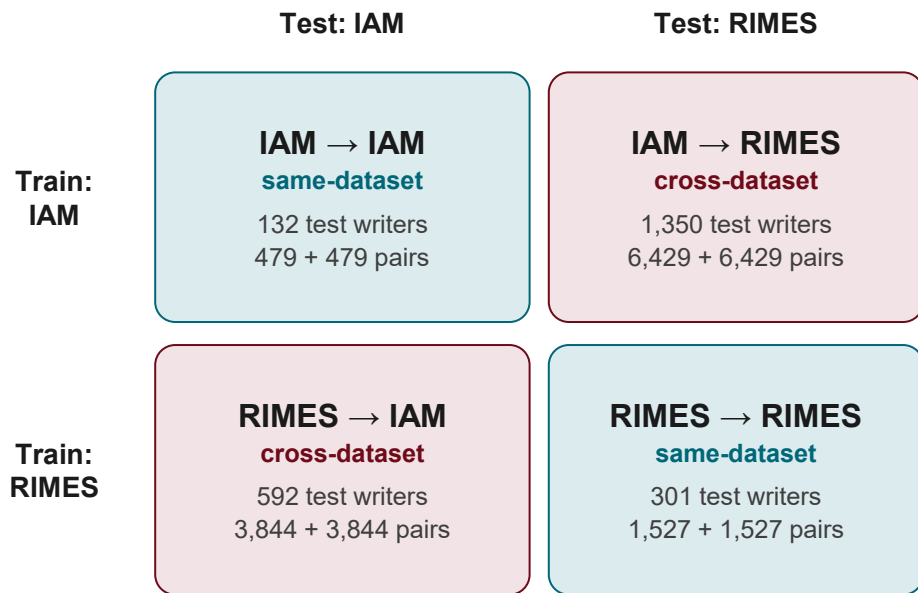
Contrastive / Triplet

noisier, non-monotonic validation loss: widening train-validation gap, most marked for Triplet

Run length

from 6 epochs (Triplet, RIMES → RIMES) to > 40 (BCE, IAM → IAM)

Writer-disjoint evaluation protocol



Same-dataset

writer-disjoint **70 / 10 / 20 %** split (seed 42)

Cross-dataset

train on **all** source writers; target split 10 % val / 90 % test

Pairs

balanced genuine / impostor test pairs, identical for all 18 configs of a split

No leakage

split by **writer**, not by image, across train, validation and test



Metrics: the biometric vocabulary

	Accept	Reject
Genuine	GA genuine accept	FR false reject
Impostor	FA false accept	GR genuine reject

$$\text{FRR} = \text{FR} / (\text{GA} + \text{FR}) \quad \text{FAR} = \text{FA} / (\text{FA} + \text{GR})$$

Not plain accuracy: accuracy / precision / recall / F1 only at the EER threshold.

EER

threshold where FAR = FRR: single-number, threshold-independent

ROC / AUC

performance across all thresholds

DET

FRR vs FAR on log-log axes

GAR @ FAR

genuine acceptance at FAR 1 % and 0.1 %: the regime of deployed systems

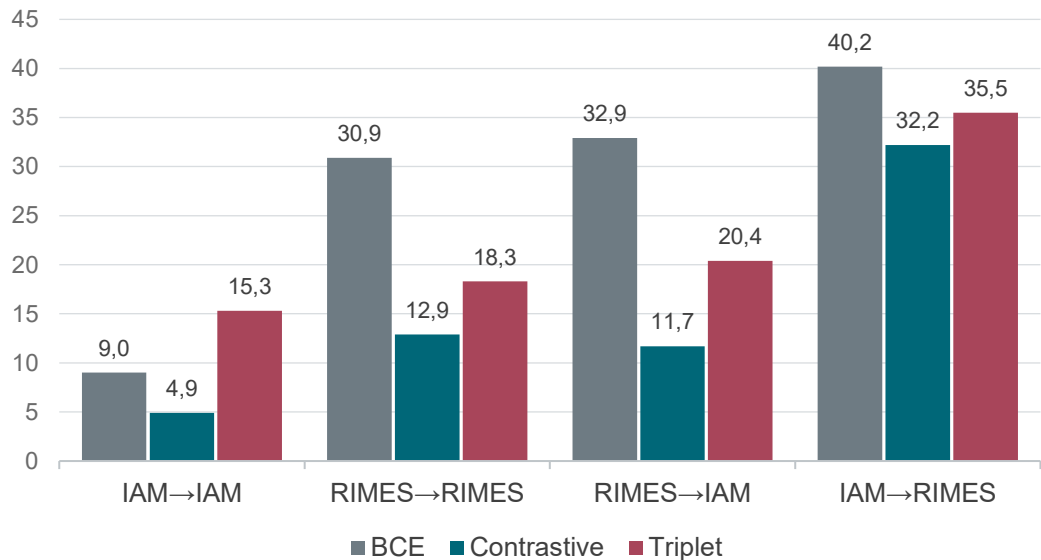
d' decidability

separation between genuine and impostor score distributions



Mean EER: Contrastive wins everywhere

Mean EER (%) across 6 backbones · lower is better



4.9 %

mean EER, Contrastive on IAM → IAM

±1.1–2.4

Contrastive EER std across backbones: the smallest variance

±15.3

BCE EER std on RIMES → RIMES: collapsed models

Same-dataset vs cross-dataset

Split	BCE			Contrastive			Triplet		
	AUC	EER %	d'	AUC	EER %	d'	AUC	EER %	d'
IAM → IAM	0.956	9.0	3.10	0.990	4.9	3.64	0.924	15.3	1.84
RIMES → RIMES	0.734	30.9	1.03	0.927	12.9	2.12	0.888	18.3	1.69
RIMES → IAM	0.715	32.9	0.90	0.950	11.7	2.13	0.869	20.4	1.31
IAM → RIMES	0.621	40.2	0.50	0.738	32.2	0.77	0.699	35.5	0.64

Mean across 6 backbones (Tables 3–4). Same-dataset rows first, then cross-dataset.

Generalisation gap

all losses drop cross-dataset; BCE has the largest AUC drop

Source writer pool

RIMES → IAM (1,500 train writers) >>
IAM → RIMES (657): AUC 0.950 vs 0.738

Close to same-dataset

Contrastive RIMES → IAM (0.950) ≈
RIMES → RIMES (0.927)



Best configuration per split

IAM → IAM

MobileNetV3-Large
+ Contrastive

0.995

AUC

EER 3.1 %

GAR @ FAR 1 % / 0.1 %
87.5 % / 72.4 %

accuracy @ EER τ : 0.969

RIMES → RIMES

ResNet-18
+ Contrastive

0.934

AUC

EER 11.5 %

GAR @ FAR 1 % / 0.1 %
53.9 % / 11.5 %

accuracy @ EER τ : 0.885

RIMES → IAM

EfficientNet-B0
+ Contrastive

0.967

AUC

EER 8.2 %

GAR @ FAR 1 % / 0.1 %
62.6 % / 13.0 %

accuracy @ EER τ : 0.918

IAM → RIMES

EfficientNet-B1
+ Contrastive

0.772

AUC

EER 29.6 %

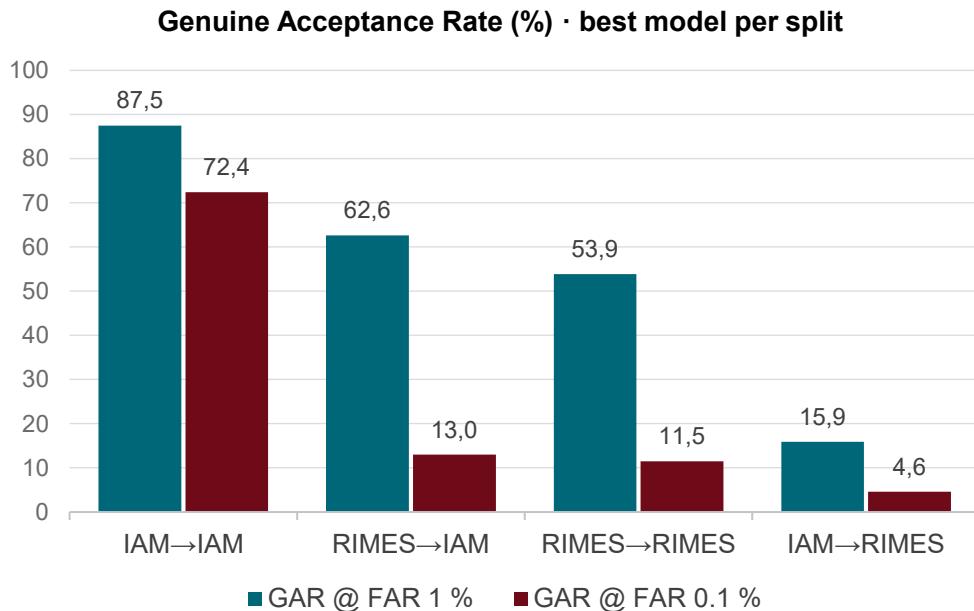
GAR @ FAR 1 % / 0.1 %
15.9 % / 4.6 %

accuracy @ EER τ : 0.704

All four winners use **Contrastive loss**; the best backbone changes with the split.



EER hides the low-FAR regime

**88.5 %**

genuine rejection rate at FAR 0.1 % (best RIMES → RIMES model)

95.4 %

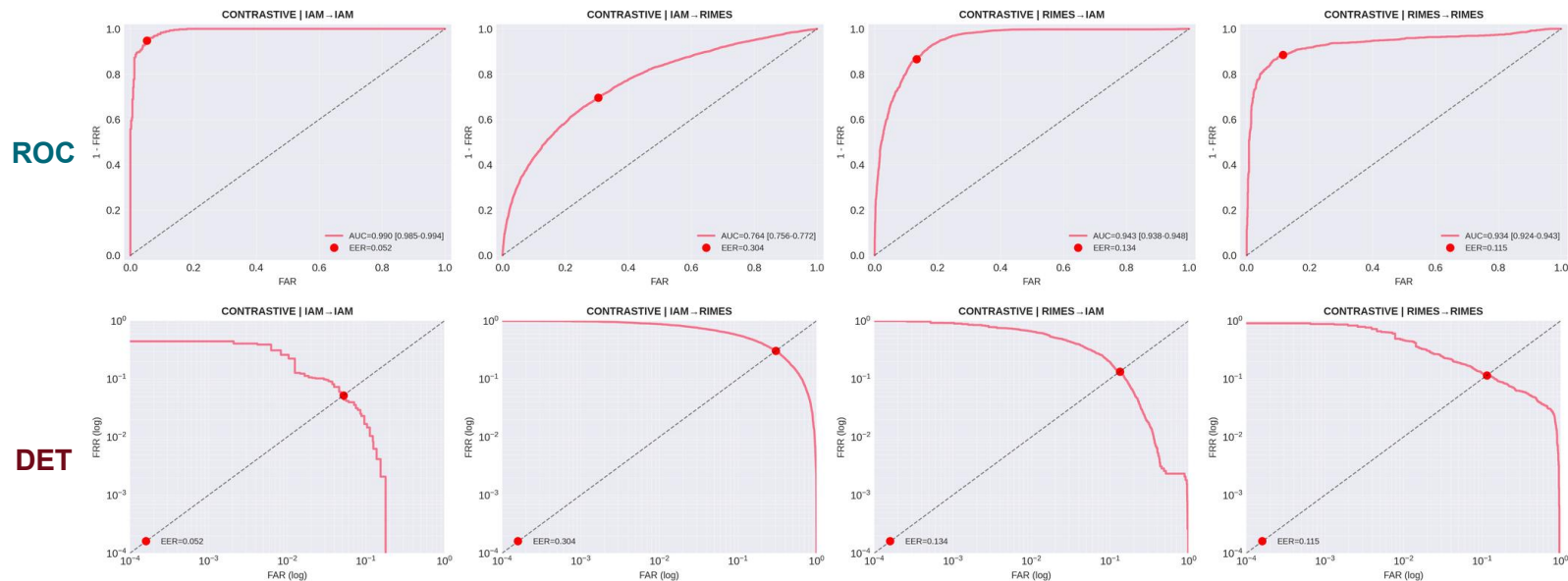
genuine rejection rate at FAR 0.1 % (best IAM → RIMES model)

Finding

Good discrimination (EER) does not imply effective operation under strict false-acceptance constraints.



ROC and DET curves



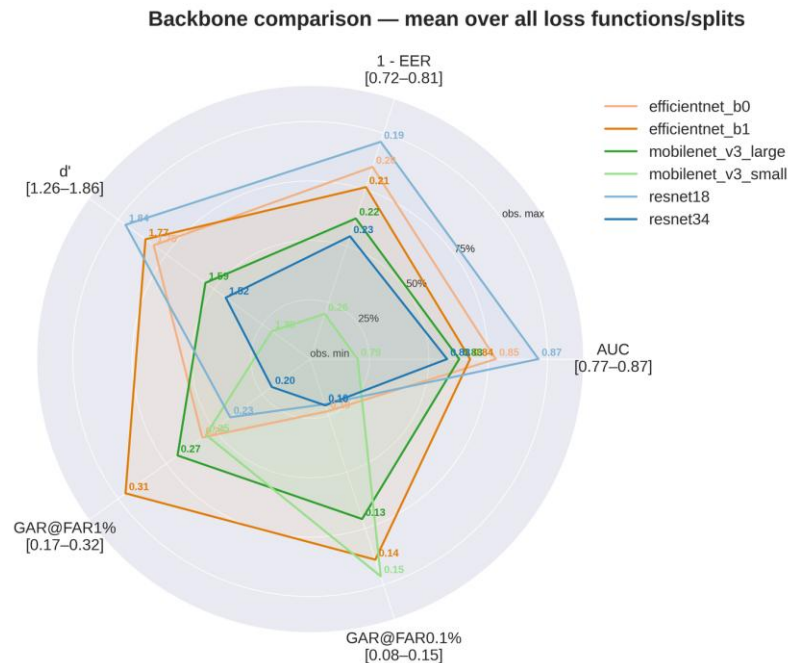
ResNet-18 + Contrastive. Same-dataset curves rise to the top-left corner; IAM $\rightarrow$ RIMES stays close to the chance diagonal.



Per-backbone profile

Backbone	AUC	EER %	d'	GAR@1%	GAR@0.1%
ResNet-18	0.869	19.2	1.84	23.3	9.8
EfficientNet-B0	0.849	20.2	1.75	25.3	10.0
EfficientNet-B1	0.838	21.1	1.77	31.0	14.3
MobileNetV3-L	0.833	22.3	1.59	27.1	13.1
ResNet-34	0.828	23.1	1.52	20.2	9.8
MobileNetV3-S	0.788	26.3	1.38	25.0	14.8

- **ResNet-18**: best AUC / EER / d', yet 2nd-lowest GAR @ FAR 1 %
- **ResNet-34** < ResNet-18 on all metrics but GAR@0.1 % (tie), 2× params
- **4 collapsed BCE models** (AUC ≈ chance): MNetV3-S ×2, -L, EffNet-B1



Mean over all losses and splits; axes scaled to observed min–max



Is the loss effect consistent?

Contrastive

>

Triplet

>

BCE**Contrastive vs BCE**

mean AUC 0.901 vs 0.756

$$p = 2.38 \cdot 10^{-7}$$

effect size $|\rho| = 0.993$ **Contrastive vs Triplet**

mean AUC 0.901 vs 0.845

$$p = 1.19 \cdot 10^{-7}$$

effect size $|\rho| = 1.000$ **Triplet vs BCE**

mean AUC 0.845 vs 0.756

$$p = 0.0115$$

effect size $|\rho| = 0.580$

Caveat: $n = 24$ (6 backbones $\times$ 4 splits), single seed, shared data: evidence of **directional consistency**, not 24 independent replicates.



Check: the full impostor pool

4,051

impostor pairs · balanced 1:1

1,179,440

impostor pairs · exhaustive pool

≈ 291×

more impostor pairs

Protocol	Impostor pairs	AUC	EER	FAR@0.1%†	GAR@0.1%	FAR@1%†	GAR@1%
Balanced	4,051	0.943	13.4 %	0.078 %	8.0 %	0.989 %	34.0 %
Exhaustive	1,179,440	0.945	13.0 %	0.100 %	9.6 %	1.000 %	34.5 %

RIMES → IAM · ResNet-18 + Contrastive (Table 8). † Effective FAR at the threshold selected for the nominal target.

In line across all metrics

No meaningful degradation: balanced 1:1 subsampling was already a reliable estimate.

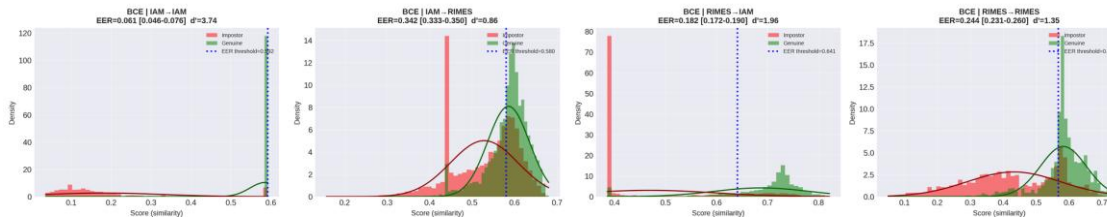
Low-FAR gap = threshold resolution

4,051 impostors cannot resolve the 0.1 % target finely; tested on one configuration only.

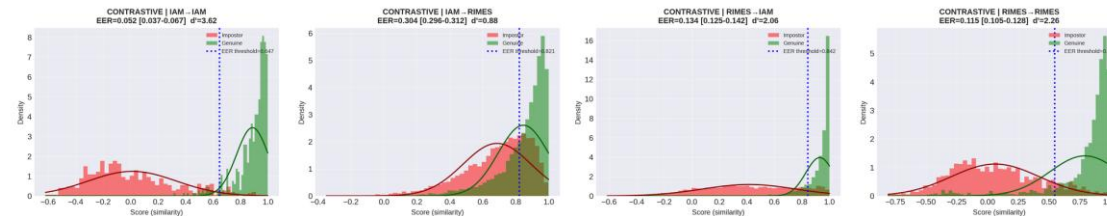


Score distributions and decidability

BCE



Contr.



ResNet-18 · green: genuine, red: impostor, dotted: EER threshold

Scores differ across losses (sigmoid [0,1] vs cosine [-1,1]): only rank-based metrics and d' are compared.

- **Non-Gaussian:** all 72 genuine + 72 impostor distributions (Shapiro–Wilk) → d' only descriptive
- BCE: isolated **impostor spikes** → possible degenerate embedding direction
- Contrastive, cross-dataset: impostors **saturate near 1** → overlap with genuine mode

Failure cases under domain shift

False accepts · impostor pairs

[illegible]

scores 0.9986 / 0.9964 vs $T = 0.8424$

False rejects · genuine pairs

label=genuine score=0.3076 (threshold=0.8424, DA=15000)
m01-131.png m01-110.png

Then can be like shell of the
flakes down on mouth independent
they respectively give us their hyper-
activation if you avoid such
premises, it may be that in our
own about forty little
phases. But in - and there is
way to tell to provide me
independents. What these interests
are, beyond personal security,
I haven't - they, but also having
in front of the important work

Wanted finally belonged to them to enter the case
give the impression of having been thought and not
detached upon inside they and then a while, while
I've passed by their to demand regularly someone
of this presence they just started at him, having
their hands like some spectators as he walked up and
down, up and down.

language score=0.2111 (threshold=0.8424, $\lambda=1.0535$) g07-038.png	g07-047.png
she felt obliged to accompany his departure. "about designating," she said from the description of his "poor brother," so she characteristically referred to him. "I had not a knowledge of him, and it would be impossible for him to have been living longer. Whether I think that she referred to the return of his family owing to this or to herself - she believed completely in the report, and she had learned from her own grandmother, the Countess of	she had heard his name to be left on your beginning from me, which, very naturally, had it seemed to me at the many minutes of his relative misfortune - the writing of the form, which she had now caused to happen, his standing position, but the return of his family his mother's friends with the eldest brother to be the "the father" but "the father" had nothing to do with the fact that people had as the subject of his death, but that the same was otherwise, not in detail.

scores $-0.3076 / -0.2111$ vs $r = 0.8424$

False accepts

Different writers, shared
slant, spacing, regularity:
style not disentangled from
identity

False rejects

Same writer, different size, pressure, legibility: multi-individuality of handwriting



Positioning against prior work

Work	Task	Data	Metric	Comparable?
Fiel & Sablatnig	closed-set identification	IAM (train), ICDAR13 / CVL	Acc. / mAP	no (task)
Christlein et al.	closed-set identification	ICDAR13, CVL	mAP	no (task)
Keglevic et al. (triplet CNN)	retrieval / identification	ICDAR13	mAP 86.1 %	no (task)
Afzali et al. (CSCNN)	verification, Siamese	IAM, IFN/ENIT	Acc., FAR, FRR	partially
Scribe verification	verification	Bamboo Slips, MCCD	AUC 0.86–0.96, EER 7.8–21 %	partially (script)
Ours · RIMES → IAM	verification, cross-dataset	RIMES → IAM	AUC 0.967, EER 8.2 %	EfficientNet-B0
Ours · IAM → IAM	verification, same-dataset	IAM	AUC 0.995, EER 3.1 %	MobileNetV3-L

Not a state-of-the-art claim: the contribution is a **controlled, uniformly-evaluated comparison** of the three objectives.





Limitations

Single seed: point estimates, no per-model bootstrap confidence intervals

Single split, no k-fold: same-dataset results may depend on the writer partition (GPU budget)

Cross-loss scores: sigmoid $[0,1]$ vs cosine $[-1,1]$: only rank metrics and d' compared

Wilcoxon non-independence: 24 observations share backbones, splits and runs

Exhaustive validation scope: a single (backbone, loss, split) triple

EER-threshold calibration: metrics computed on the same set: mildly optimistic

Limited augmentation: only resized crop (0.9–1.1) and rotation up to 15°

Non-Gaussian scores: d' / decidability used only as descriptive indices

Failure-analysis granularity: ResNet-18 only; top-2 pairs for a single configuration

Training dynamics: epoch counts not consolidated; logs inspected only for a subset



Future directions

Statistical robustness

multi-seed training +
bootstrap confidence
intervals

Broader architectures

transformer-based and
attention-augmented
backbones

Advanced negative sampling

online hard-negative mining,
embedding-distance adaptive
sampling

≈ hard triplets (Ch.6)

Fine-grained failure analysis

writer- and pair-level, across
the full test sets

Explainability

Grad-CAM on the regions
driving decisions

Embedding dimensionality

PCA intrinsic dimensionality
or output-size sweep

≈ PCA (Ch.6)

Multi-modal integration

with signatures and online
writing dynamics

≈ multimodal systems (Ch.12)

Doddington zoo analysis

per-writer biometric
menagerie classes

≈ menagerie (Ch.3)



Future work: Doddington zoo analysis

Idea

- Per-writer categorisation into biometric menagerie classes (Yager & Dunstone)

Question

- Are errors spread across writers, or concentrated in a few problematic identities?

Requirement

- Genuine and impostor scores aggregated **per writer**: not retained in the current pipeline

Link to failure cases

- Do false accepts / rejects come from a few chronically difficult, multi-style writers?

Class	Course definition
Goat	matches poorly with itself → ↑ FRR
Lamb	easy to imitate → ↑ FAR
Wolf	good at imitating others → ↑ FAR
Dove	high genuine, low impostor scores: ideal user
Chameleon	high genuine and impostor scores → FA
Phantom	low genuine and impostor scores → FR
Worm	low genuine, high impostor scores: worst case

Classes as named in the paper; definitions from the course (Ch.3).



Take-home messages

1

$p < 10^{-6}$

Metric learning beats BCE

Both metric losses outperform BCE;
Contrastive is the most consistent
across backbones and splits.

2

3.1 % → 29.6 %

**Generalisation is the real
challenge**

Best EER within IAM vs best IAM →
RIMES; source writer-pool size is a
contributing factor.

3

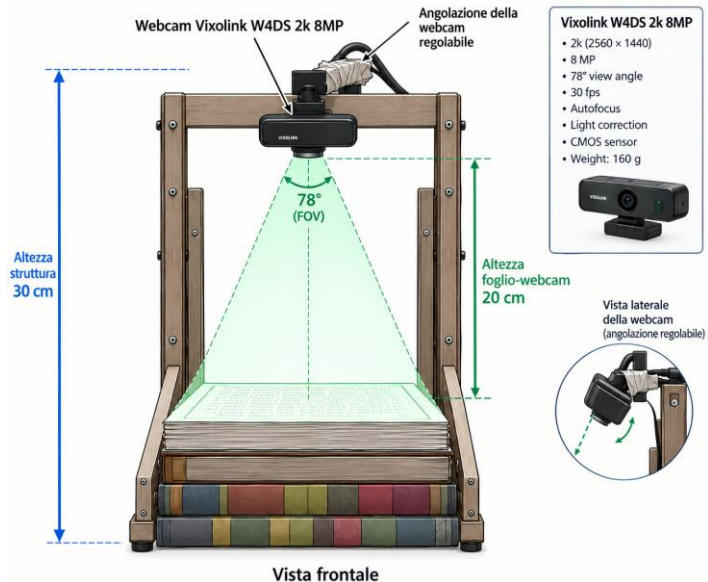
4.6 %

Report operating points

GAR at FAR = 0.1 % for the best
IAM → RIMES model: a low EER
does not imply usable operation.



Demo: acquisition setup



Hardware

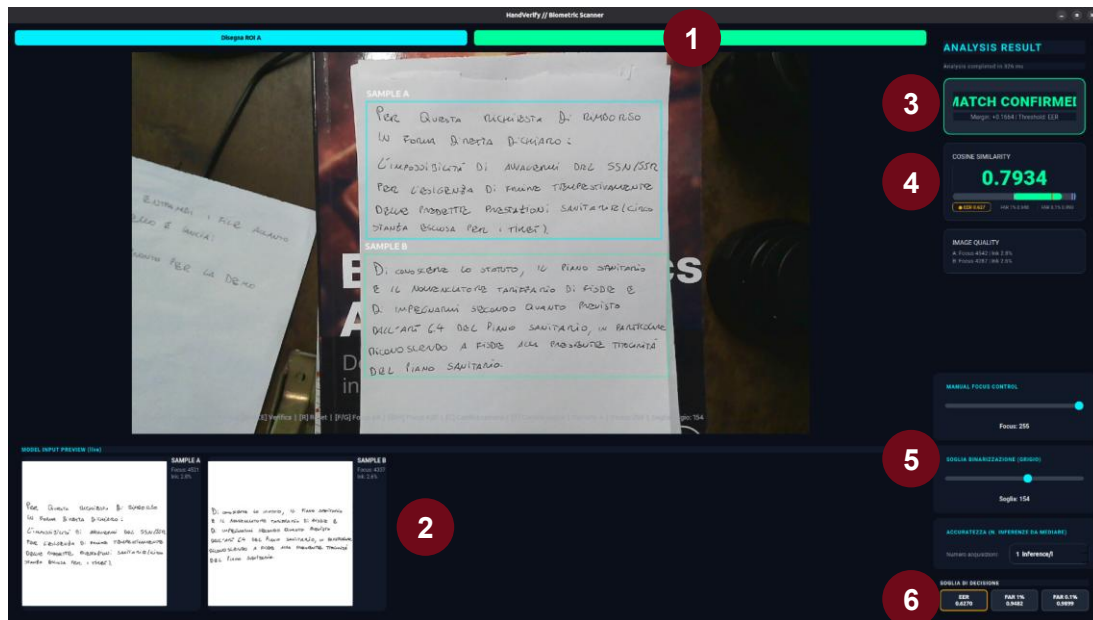
- Webcam Vixolink W4DS
- 2K (2560×1440) · 8 MP
- 78° field of view · 30 fps

Geometry

- Frame height ≈ 30 cm
- Sheet-webcam distance ≈ 20 cm
- Adjustable webcam angle, top-down view



Demo: how the interface works

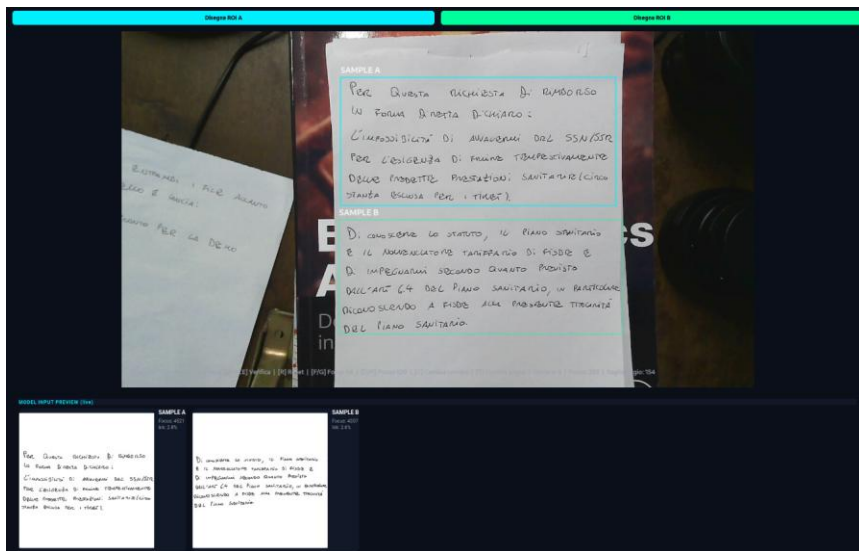


- 1 **Draw ROI A / B**
drag on the live video around each sample
- 2 **Model input preview**
live 448x448 inputs, with focus and ink
- 3 **Analysis result**
MATCH / NO MATCH, margin, threshold
- 4 **Cosine similarity**
score bar with the 3 threshold markers
- 5 **Controls**
focus, grey binarisation, N inferences averaged
- 6 **Decision threshold**
EER · FAR 1 % · FAR 0.1 %

SPACE verify · R reset · T threshold



Demo: same writer, different text



Live ROIs A / B and model input preview (448×448)



Analysis result

MATCH

0.7934

cosine similarity

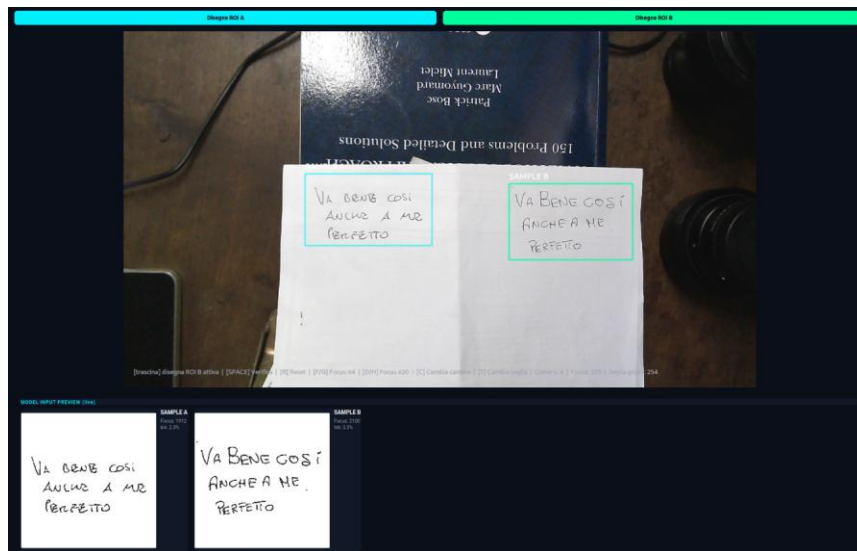
0.6270

EER threshold

+0.1664

margin

Demo: same text, different writers



Live ROIs A / B and model input preview (448x448)



Analysis result

NO MATCH

0.5616

cosine similarity

0.7912

EER threshold

-0.2296

margin



References

- Bromley et al., Signature verification using a “Siamese” TDNN, NIPS 1993
- Hadsell, Chopra, LeCun, Dimensionality reduction by learning an invariant mapping, CVPR 2006
- Schroff, Kalenichenko, Philbin, FaceNet, CVPR 2015 · Koch et al., Siamese one-shot, ICML-W 2015
- He et al., ResNet, CVPR 2016 · Tan & Le, EfficientNet, ICML 2019 · Howard et al., MobileNetV3, ICCV 2019
- Deng et al., ImageNet, CVPR 2009 · Ioffe & Szegedy, Batch normalization, ICML 2015
- Bensefia et al., PRL 2005 · Said et al., PR 2000 · Schomaker & Bulacu, TPAMI 2004
- Bulacu & Schomaker, TPAMI 2007 · Plamondon & Srihari, TPAMI 2000
- Fiel & Sablatnig, CAIP 2015 · Christlein et al., GCPR 2015 / ICDAR 2017 · He & Schomaker, FragNet, TIFS 2020
- Keglevic, Fiel, Sablatnig, Triplet CNNs for writer retrieval, ICFHR 2018
- IAM Handwriting Database (FKI) · RIMES-Lines dataset (Kaggle)
- Mansfield & Wayman, Best practices in biometric testing, NPL 2002
- He & Garcia, Learning from imbalanced data, TKDE 2009 · Loshchilov & Hutter, AdamW, ICLR 2019
- Rosenblum et al., Incorrect statistical reasoning ... forensic evidence, PNAS 2024
- Moszczynski, The multi-individuality of handwriting, FSI 2019
- Afzali et al., IJRIE 2024 · Wang et al., Bamboo slip scribe verification, npj Herit. Sci. 2026
- Yager & Dunstone, The biometric menagerie, TPAMI 2010
- M. De Marsico, Biometric Systems course slides, Sapienza University of Rome
- Slide template: github.com/pietro-nardelli/sapienza-ppt-template (CC BY-NC-SA 4.0)





Thank you for the attention!



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